



ALPHA ARTS AND SCIENCE COLLEGE

PORUR, CHENNAI-600 116

A PROJECT REPORT ON

**LEGALEASE – AI-POWERED LEGAL
DOCUMENT GENERATOR**

SUBMITTED BY

TEAM LEAD:

❖ **SAAI SRI B**

AA0412C7D021A67B5DA42831A098AE66

TEAM MEMBERS:

❖ **PRADEEPA. S**

47522F570609407880C46A6E13D555A2

❖ **RENUGA. V**

4DBC4B21BCA8F5D9B774AC7C5EB4EB0B

❖ **THEJALINI. S**

BCFC8A2787167D5AB934E939118EA9BE

❖ **JAYAKIRUBA. D**

BD5F43E555EA311808D45FA254443FB7

1 INTRODUCTION

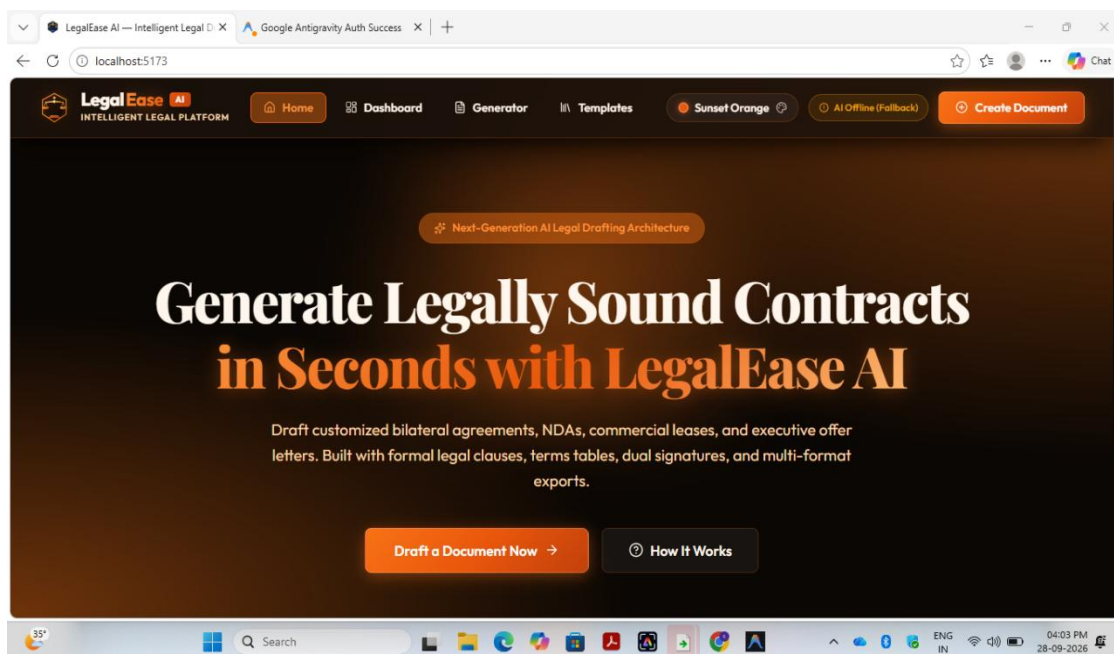
1.1 Project Overview

Legalese is an AI-powered web application designed to simplify the creation of legal documents. The system uses Generative Artificial Intelligence to generate customizable and editable legal documents based on information provided by the user.

The application can generate different types of documents such as:

- Employment Contracts
- Lease Agreements
- Non-Disclosure Agreements (NDA)
- Freelance Contracts
- General Agreements
- Employment Offer Letters

The user provides important details such as the document type, parties involved, terms and conditions, and effective date. Based on these inputs, the AI generates a structured legal document.



Objectives

The main objectives of Legalease are:

1. To develop an AI-based legal document generation system.
2. To generate documents based on user-provided information.

3. To provide an easy-to-use web interface.
4. To allow users to edit generated content.
5. To provide professional document formatting.
6. To support TXT, DOCX, and PDF downloads.
7. To integrate Google's Gemini AI for content generation.
8. To provide a simple solution for users without advanced legal knowledge.

2 – SYSTEM ANALYSIS AND TECHNOLOGIES

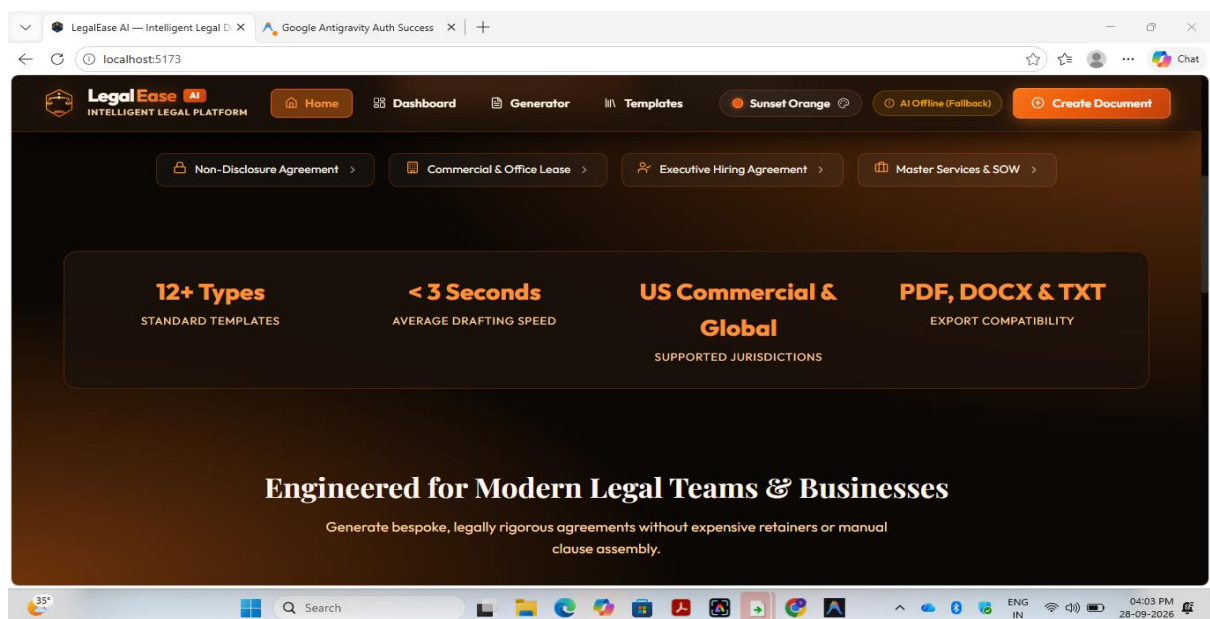
2.1 Existing System

In the traditional approach, legal documents are generally created manually using word-processing software or existing templates.

Users need to:

- Search for an appropriate template.
- Enter the required information.
- Modify clauses manually.
- Format the document.
- Check the document structure.
- Save or convert the document into the required format.

This process can require considerable time and knowledge of legal-document formatting.



2.2 Proposed System

Legalease provides an AI-assisted approach where users enter their requirements through a web interface. The application sends these requirements to the backend, which communicates with the Gemini AI model.

The generated document is then returned to the frontend where the user can:

Enter Details → Generate Document → Preview → Edit → Download

The proposed architecture consists of four major components:

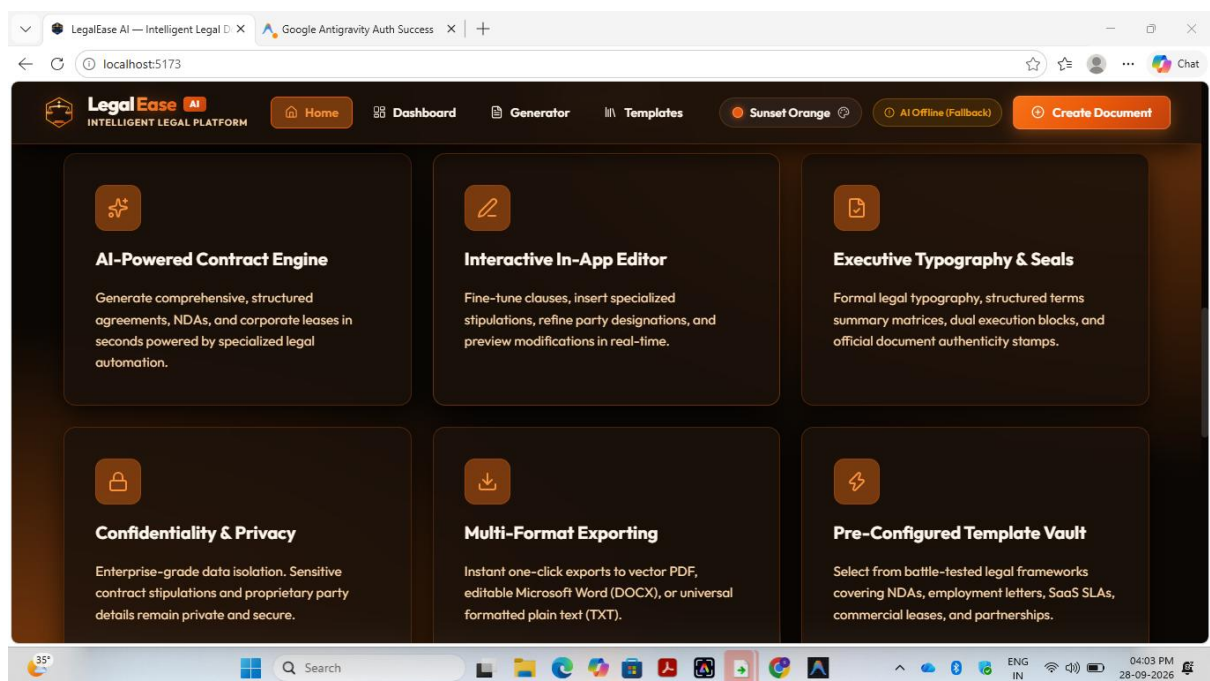
1. **Frontend – Streamlit**
2. **Backend – FastAPI**
3. **AI Core – Gemini**
4. **Document Formatting Modules – DOCX, PDF and TXT**

The original project documentation identifies Streamlit for user interaction, FastAPI for handling API requests, Gemini for content generation, and formatting modules for document export.

PAGE 3 – SYSTEM ARCHITECTURE AND MODULES

3.1 System Architecture

The frontend collects document type, parties, terms, and effective date. The backend validates and processes these inputs before sending them to the AI component.



3.2 Frontend Module

The frontend is developed using Stream lit.

It provides fields for:

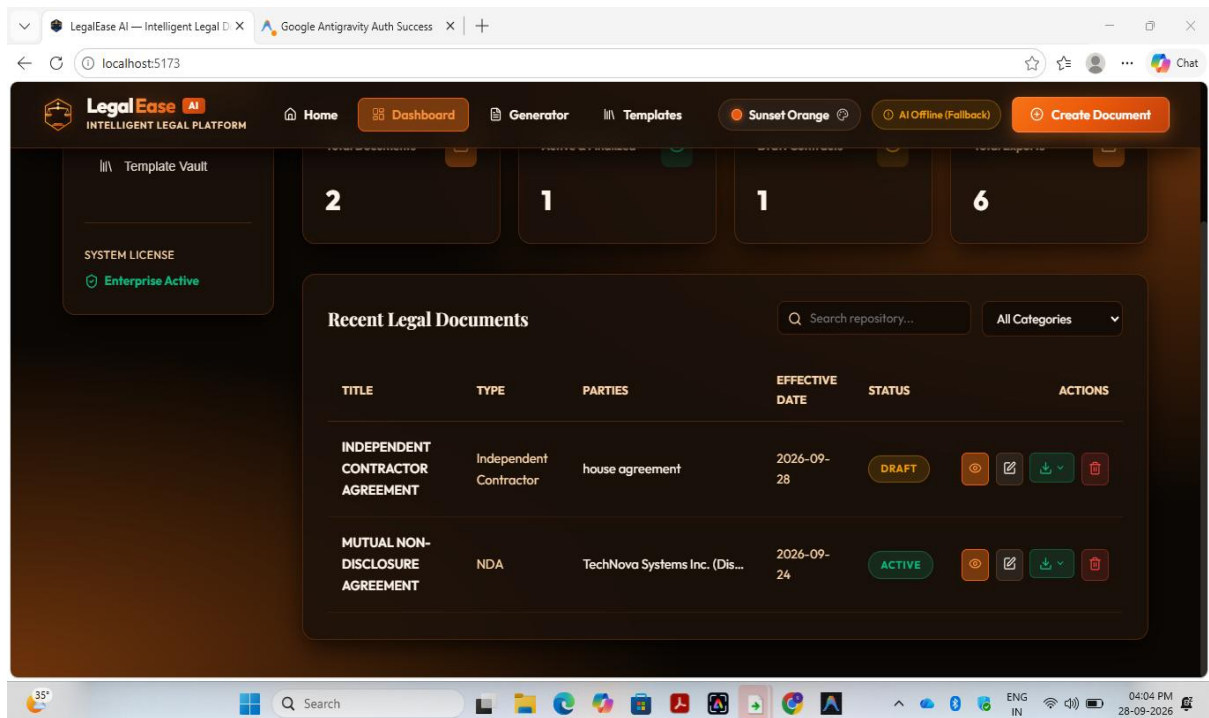
- Document Type
- Parties Involved
- Terms and Conditions
- Effective Date

3.3 Backend Module

The backend is developed using Fast API.

The main responsibilities are:

- Receiving frontend requests.
- Validating input.
- Creating the document request.
- Calling the Gemini generator.
- Returning generated content.



3.4 AI Generation Module

The AI module communicates with Gemini.

The process is:

User Input



Structured Prompt



Gemini AI



Generated Legal Content



Fast API



Stream lit

The AI-generated output is then formatted and presented to the user.

4 – IMPLEMENTATION AND FUNCTIONALITIES

4.1 Document Generation

The user first selects the required document type.

For example:

Document Type: Freelance Work Contract

Then the user provides the parties involved.

Parties:

Jane Doe – Service Provider

Tech Nova Inc. – Client

The user can then enter terms and conditions such as:

- Payment within 30 days.
- Work completion according to the agreed deadline.
- Confidentiality requirements.
- Termination conditions.

The effective date is also provided.

4.2 Document Preview

After the user selects **Generate Document**, the application sends the information to the backend.

The generated content is displayed as a styled preview. The user can review the document before downloading it.

TXT Export

The generated document can be saved as a plain-text .txt file.

This provides a simple format that can be opened using basic text editors.

4.5 DOCX Export

The system uses python-docx to create Microsoft Word documents.

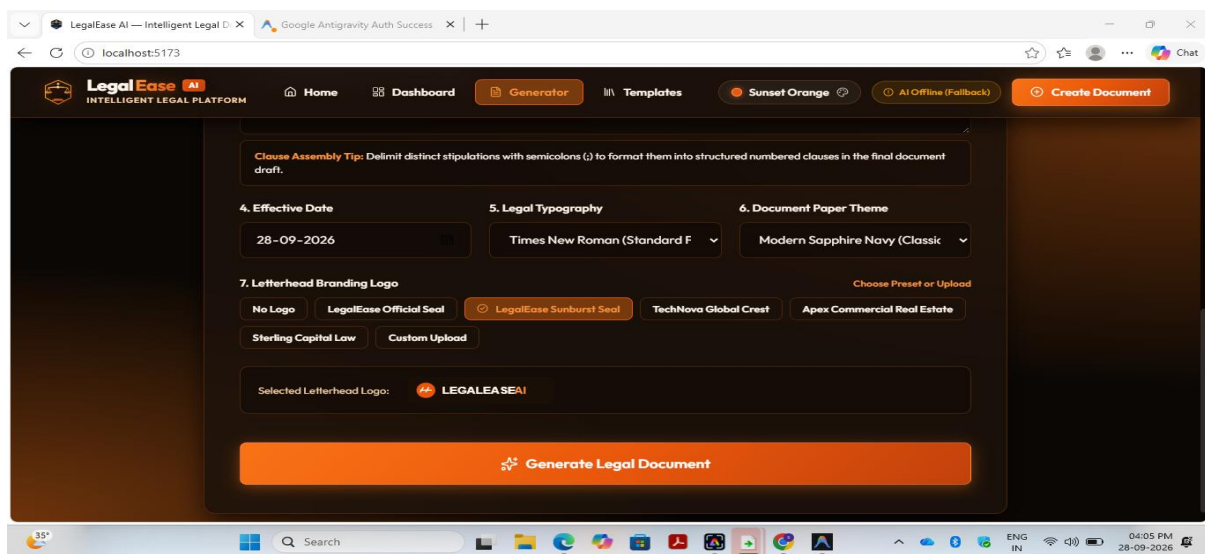
The DOCX output can contain:

- Logo
- Document title
- Headings
- Paragraph formatting
- Terms table
- Footer

Security and Configuration

The Gemini API key is stored using an environment configuration file rather than directly placing the key inside the source code.

A .env file is used for this purpose.



5 – TESTING, ADVANTAGES, FUTURE ENHANCEMENTS AND CONCLUSION

5.1 Testing

The application should be tested using different combinations of:

- Document types.
- Parties.
- Terms and conditions.
- Effective dates.

Important areas to verify include:

5.2 Advantages

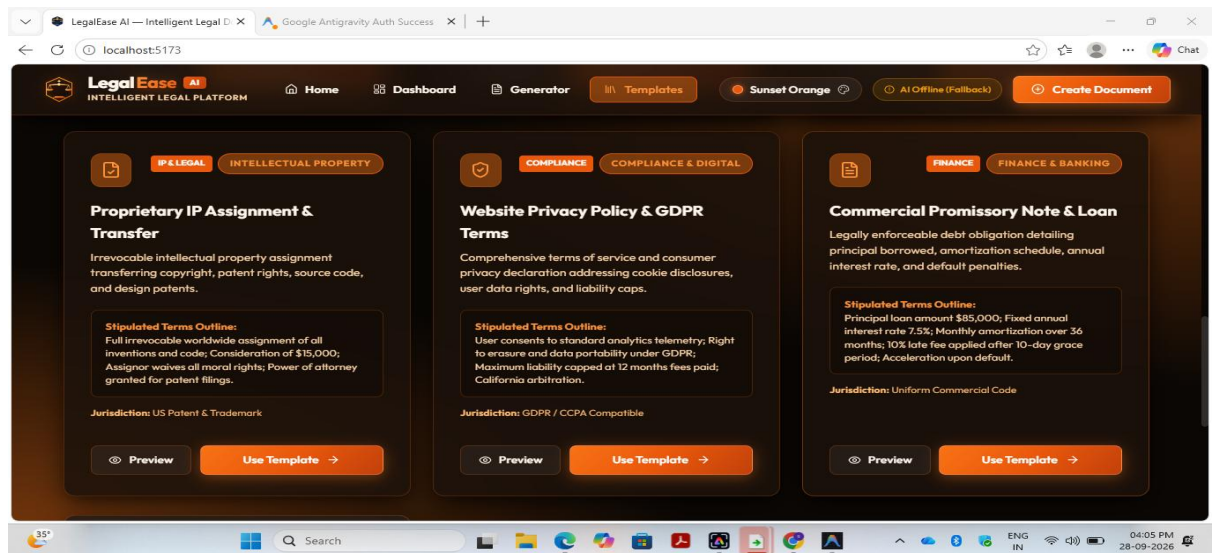
The major advantages of Legalease are:

- AI-assisted document generation.
- Easy-to-use interface.
- Faster preparation of standard legal documents.
- Editable generated content.
- Multiple download formats.
- Professional document formatting.
- Support for branding elements such as logos.
- Integration between frontend, backend, and AI services.

5.3 Limitations

Some limitations of the system include:

- AI-generated legal content should be reviewed before actual legal use.
- The quality of output depends on the information supplied by the user.
- An internet connection and valid AI API configuration are required for AI generation.
- The system is intended for document generation and does not replace professional legal advice.



Future Enhancements

Possible future improvements include:

- More legal-document templates.
- Multilingual document generation.
- Advanced contract analysis.
- Integration with legal databases.
- User accounts and document history.
- Cloud-based deployment.
- Improved document customization.
- Additional AI-powered clause analysis.

The uploaded documentation also identifies deeper contract analysis, legal-database integration, and personalized recommendations as possible future enhancements.

5.5 Conclusion

Legalese demonstrates how Generative AI can be integrated with modern web technologies to simplify the preparation of structured legal documents. The combination of **Stream lit**, **Fast API**, **Gemini AI**, **Python-docx**, and **FPDF** provides an end-to-end workflow from user input to document generation and export.

The system allows users to enter their requirements, generate AI-assisted legal content, preview and edit the document, and download the result in multiple formats. Overall, the project demonstrates the practical application of Generative AI, API development, document automation, and user-interface design in a single software solution.

GitHub link: <https://github.com/saaisribala03-netizen/legalease.git>

Demo video link:

<https://drive.google.com/file/d/1648ob5Alct6W2NwFCmhCku2LivjrmhWA/view?usp=drivesdk>